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Aryne-mediated [2,3]-sigmatropic rearrangement of tertiary 2,3-allenylamines bearing an electron-withdrawing group at the α-position†

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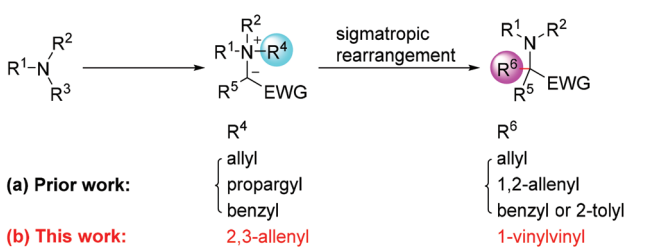
An unprecedented [2,3]-sigmatropic rearrangement reaction of quaternary 2,3-allenylammonium ylides, generated *in situ* from tertiary 2,3-allenylamines and arynes, has been established. With 2-(trimethylsilyl) aryl triflates as aryne precursors, a range of tertiary 2,3-allenylamines bearing an electron-withdrawing group at the α-position smoothly participated in the aryne-mediated [2,3]-sigmatropic rearrangement at room temperature, delivering structurally diverse 2-vinylallyamines or 1-amino-1,3-dienes in moderate to excellent yields. The reaction proceeds in the absence of strong bases and transition metals, is compatible with moisture and air, and tolerates a wide variety of functional groups.

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Introduction

Owing to charge acceleration, transformation of tertiary amines into the corresponding quaternary ammonium ylides, bearing an electron-withdrawing group at the α-position, significantly reduces the reaction temperature for the sigmatropic rearrangements *via* C–N bond cleavage, permitting the process to serve as a key step in the construction of complex nitrogen-containing molecules.[1] In this regard, a number of quaternary allyl-, propargyl-, and benzylammonium ylides have been reported to be prepared *via* stepwise or one-pot quaternization of tertiary amines followed by deprotonation, and participate in a [2,3]- or a [1,2]-sigmatropic rearrangement to afford functionalized tertiary amines that are not readily synthesized by alternative methods (Scheme 1a). Nevertheless, to our knowledge, it remains unexplored for the sigmatropic rearrangement of quaternary 2,3-allenylammonium ylides, which would yield 2-vinylallylamines as promising versatile building blocks in chemical synthesis through transformations such as oxidation, carbonylation, and cycloaddition (Scheme 1b).[2]

Arynes have recently emerged as powerful electrophiles to activate tertiary allyl-,[3] propargyl-,[4] and benzylamines[5] bearing an electron-withdrawing group at the α-position under mild conditions to generate *in situ* the corresponding quaternary ammonium ylides,[6] which participate in sigmatropic rearrangements to produce various aniline derivatives. In contrast to the *in situ* generation of quaternary ammonium ylides *via* transition metal-catalyzed *N*-alkylation of tertiary allyl-[7] or propargylamines with diazo compounds[8] or allylic carbonates[9] or *via* zinc halide-catalyzed ring-opening of epoxides by tertiary amines,[10] the aryne-mediated sigmatropic rearrangements of tertiary amines avoids the employment of transition metals, and moreover, introduces *N*-aryl groups into the target molecules. In this context, we envisioned that similar addition of tertiary 2,3-allenylamines bearing an electron-withdrawing group at the α-position to arynes followed by proton transfer would lead to the formation of quaternary 2,3-allenylammonium ylides, which would undergo a [2,3]-sigmatropic rearrangement to yield 2-vinylallyamines (Scheme 2). Care should be taken to prevent the aryl group from participating in an aromatic aza-Claisen rearrangement.[11] In exposure to a base, some of the 2-vinylallyamine products would further isomerize to 1-amino-1,3-dienes.[4a]

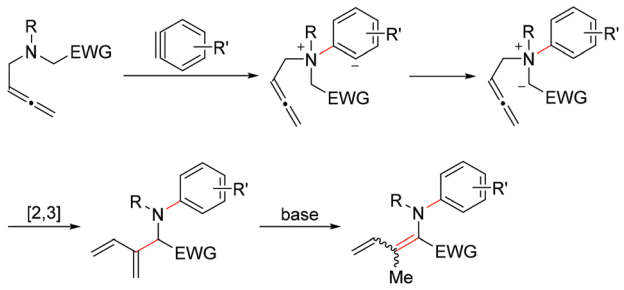


Scheme 1 Sigmatropic rearrangements of quaternary ammonium ylides. EWG = electron-withdrawing group.

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†Electronic supplementary information (ESI) available: Experimental procedures, characterization data, and copies of NMR spectra and HPLC traces for all compounds. See DOI: 10.1039/d1ob00887k

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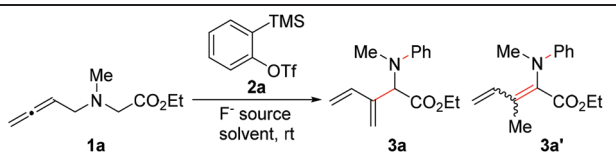


Scheme 2 Design of a new [2,3]-sigmatropic rearrangement.

Results and discussion

In the model reaction of *N*-(2,3-allenyl)glycine ester **1a** with benzyne in acetonitrile at room temperature without extrusion of moisture and air, we surveyed a few readily available fluoride sources for the *in situ* generation of benzyne *via* fluoride-induced 1,2-elimination of 2-(trimethylsilyl)phenyl triflate (**2a**) (Table 1, entries 1–8).[12] The use of TBAF predominantly afforded 1-amino-1,3-diene **3a′** (**3a** : **3a′** = 2 : 98) as a 53 : 47 mixture of *Z*- and *E*-isomers (Table 1, entry 6), and in sharp contrast, combined use of KF and 18-crown-6 afforded 2-vinylallyamine **3a** as a major isomer (**3a** : **3a′** = 84 : 16) (Table 1, entry 8). Obviously, isomerization of 2-vinylallyamine **3a** to 1-amino-1,3-diene **3a′** occurred under the weak basic conditions.[4a] In these cases, the phenyl group was not observed to participate in an aromatic aza-Claisen rearrangement. We then examined a number of solvents (Table 1, entries 9–14), and to our delight, found that performing the reaction in tetrahydrofuran enhanced the ratio between 2-vinylallyamine **3a** and 1-amino-1,3-diene **3a′** from 84 : 16 to 96 : 4 as well as the combined yield from 70% to 79% (Table 1, entry 9). Moreover, significant isomerization of 2-vinylallyamine **3a** to 1-amino-1,3-diene **3a′** was observed after prolonging the reaction time from 3 h to 20 h (Table 1, entry 15).

Table 1 Optimization of the reaction conditions[a]



Entry	F⁻ source	Solvent	Yield[b] (%)	3a : 3a′[c]
1	CsF	MeCN	34	83 : 17
2	KF	MeCN	Trace	—
3	ZnF_2	MeCN	0	—
4	NaF	MeCN	0	—
5	NH_4F	MeCN	Trace	—
6	TBAF[d]	MeCN	82	2 : 98
7	CsF/18-crown-6	MeCN	65	30 : 70
8	KF/18-crown-6	MeCN	70	84 : 16
9	KF/18-crown-6	THF	79	96 : 4
10	KF/18-crown-6	DMF	40	<1 : 99
11	KF/18-crown-6	1,4-Dioxane	57	97 : 3
12	KF/18-crown-6	Toluene	37	92 : 8
13	KF/18-crown-6	DCE	68	92 : 8
14	KF/18-crown-6	$EtOCH_2CH_2OEt$	70	97 : 3
15[e]	KF/18-crown-6	THF	76	84 : 16

[a] Reaction conditions: **1a** (0.20 mmol), **2a** (0.24 mmol), F⁻ source (0.40 mmol), 18-crown-6 (if any, 0.40 mmol), solvent (1.0 mL), rt, 3 h. [b] Combined isolated yield of **3a** and **3a′**. [c] The ratios between **3a** and **3a′** were determined by ^{1}H NMR spectroscopic analysis of the crude products, and the *Z*/*E* ratios of **3a′** were determined to be nearly 1 : 1 in all the cases. [d] A solution of TBAF (1.0 M) in THF was used. [e] The reaction was run for 20 h.

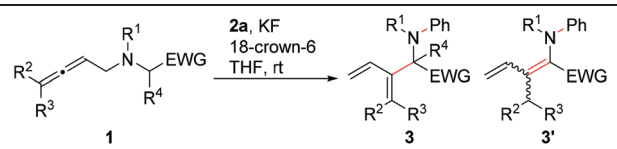
Under the optimized conditions, we observed broad substrate scope for the tertiary 2,3-allenylamines in the [2,3]-sigmatropic rearrangement reaction using 2-(trimethylsilyl) phenyl triflate (**2a**) as a benzyne precursor (Table 2). In addition to ester groups, the *p*-nitrophenyl group served as an electron-withdrawing group in the benzyne-mediated [2,3]-sigmatropic rearrangement of a tertiary 2,3-allenylamine albeit the resulting 2-vinylallyamine **3d** was isolated in a moderate yield (Table 2, entry 4). Replacing the ester group with an amide group did not lead to the formation of the desired rearrangement product (Table 2, entry 5).[13] When the reaction was extended to a tertiary 2,3-allenylamine bearing a ketone or a cyano group, the 1-amino-1,3-diene product rather than the 2-vinylallyamine product was obtained due to ready isomerization under the standard conditions (Table 2, entries 6–8). While the *N*-(2,3-allenyl)glycine ester bearing an *N*-alkyl group participated in the benzyne-mediated [2,3]-sigmatropic rearrangement to afford the corresponding 2-vinylallyamines in good yields, the reaction of an *N*-phenyl-containing analogue gave a complex mixture and no rearrangement product was isolated (Table 2, entries 9–11). The rearrangement reaction worked well with the *N*-(2,3-allenyl)glycine esters bearing a disubstituted or a trisubstituted allenyl group, but significant erosion of enantiopurity was observed in the axial-to-central chirality transfer (Table 2, entries 12–16). In contrast, L-proline-derived 2,3-allenylamine **1q** participated in the benzyne-mediated [2,3]-sigmatropic rearrangement, constructing a nitrogen-substituted quaternary stereocenter with excellent retention of enantiopurity (Table 2, entry 17). In this case, high efficiency of central-to-central chirality transfer should be attributable to the highly diastereoselective attack of the cyclic tertiary amine on benzyne.

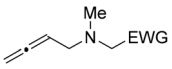
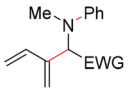
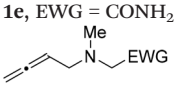
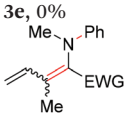
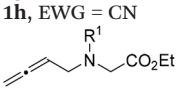
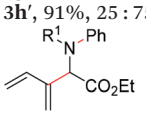
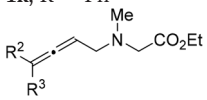
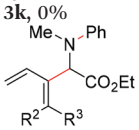
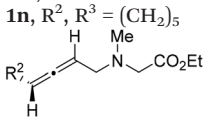
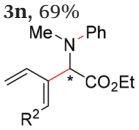
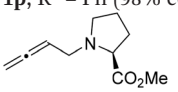
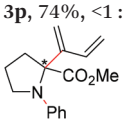
We further investigated the migratory tendency of allyl, propargyl, and benzyl relative to 2,3-allenyl in the benzyne-mediated sigmatropic rearrangements of tertiary amines under the standard conditions. In the reaction of *N*-(2,3-allenyl)-*N*-allylglycine ester **1r** with benzyne, both of 2,3-allenyl and allyl participated in the [2,3]-sigmatropic rearrangement to afford 2-vinylallyamine **3r** and homoallylamine **4a**, respectively, in a ratio of 43 : 57 (Scheme 3a). In contrast, the propargyl group was kept untouched in the benzyne-mediated [2,3]-sigmatropic rearrangement of *N*-(2,3-allenyl)-*N*-propargylglycine ester **1s** under the standard conditions, probably owing to unfavourable formation of an envelope-like transition state from a linear propargyl group relative to a 2,3-allenyl group (Scheme 3b).[1] Finally, benzyl was found to be reluctant to migrate relative to 2,3-allenyl, and only 20% of *N*-(2,3-allenyl)-

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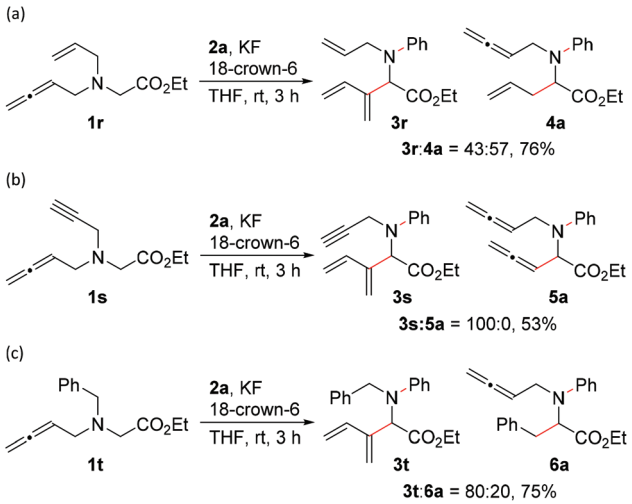
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Table 2 Scope of 2,3-allenylamines[a]



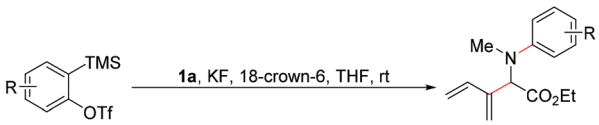
Entry	Amine	Product
		
1	**1a**, EWG = CO_2Et	**3a**, 76% (78%)[b]
2	**1b**, EWG = $CO_2{}^tBu$	**3b**, 65%
3	**1c**, EWG = CO_2Bn	**3c**, 73%
4	**1d**, EWG = 4-$O_2NC_6H_4$	**3d**, 48%
5	**1e**, EWG = $CONH_2$	**3e**, 0%
		
6	**1f**, EWG = COMe	**3f′**, 49%, >99 : 1 *Z*/*E*
7	**1g**, EWG = COPh	**3g′**, 44%, >99 : 1 *Z*/*E*
8	**1h**, EWG = CN	**3h′**, 91%, 25 : 75 *Z*/*E*
		
9	**1i**, $R^1 = (CH_2)_3Me$	**3i**, 73%
10	**1j**, $R^1 = CH_2CH{=}C{=}CH_2$	**3j**, 65%
11	**1k**, R^1 = Ph	**3k**, 0%
		
12	**1l**, $R^2 = n$-C_7H_{15}, R^3 = H	**3l**, 90%, 10 : 90 *Z*/*E*
13	**1m**, $R^2 = R^3$ = Et	**3m**, 63%
14	**1n**, R^2, $R^3 = (CH_2)_5$	**3n**, 69%
		
15	**1o**, R^2 = Cy (>99% ee)	**3o**, 85%, <1 : 99 *Z*/*E*, 70% ee
16	**1p**, R^2 = Ph (98% ee)	**3p**, 74%, <1 : 99 *Z*/*E*, 81% ee
		
17	**1q** (>99% ee)	**3q**, 39%, 90% ee

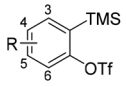
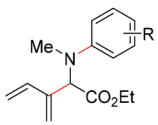
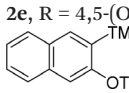
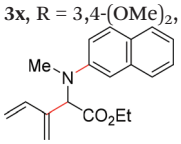
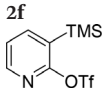
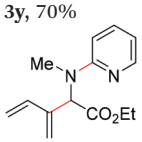
[a] Reaction conditions: **1** (0.20 mmol), **2a** (0.24 mmol), KF (0.40 mmol), 18-crown-6 (0.40 mmol), THF (1.0 mL), rt, 3 h. Isolated yields are given. The *Z*/*E* ratios were determined by ^{1}H NMR spectroscopic analysis of the crude products. The ee values were determined by HPLC analysis on a chiral stationary phase. [b] The reaction was run on a 6 mmol scale.



Scheme 3 Competing sigmatropic rearrangements.

Table 3 Scope of arynes[a]



Entry	Aryne precursor	Product
		
1	**2b**, R = 6-Me	**3u**, R = 3-Me, 92%
2	**2c**, R = 4-Me	**3v** : **3u** (R = 4-Me) = 45 : 55,[b] 60%
3	**2d**, R = 4,5-Me_2	**3w**, R = 3,4-Me_2, 72%
4	**2e**, R = 4,5-$(OMe)_2$	**3x**, R = 3,4-$(OMe)_2$, 76%
		
5	**2f**	**3y**, 70%
		
6	**2g**	**3z**, 0%

[a] Reaction conditions: **1a** (0.20 mmol), **2** (0.24 mmol), KF (0.40 mmol), 18-crown-6 (0.40 mmol), THF (1.0 mL), rt, 3 h. Isolated yields are given. [b] Determined by ^{1}H NMR spectroscopic analysis of the crude product.

N-benzylglycine ester **1t** participated in the benzyl [1,2]-sigmatropic rearrangement (Scheme 3c).

A few aryne precursors, 2-(trimethylsilyl)aryl triflates, were examined with regard to their potential in promoting the [2,3]-sigmatropic rearrangement of *N*-(2,3-allenyl)glycine ester **1a** (Table 3). In the reaction of an *in situ* generated unsymmetrical aryne, the regioselectivity highly depends on the position of substituents. In line with previous reports,[14] a single regioisomer was obtained from the reaction of 3-methylbenzyne (Table 3, entry 1), and the amine addition to 4-methylbenzyne exhibited poor regioselectivity (Table 3, entry 2). A few symmetrical arynes were found to mediate the [2,3]-sigmatropic rearrangement of *N*-(2,3-allenyl)glycine ester **1a**, delivering the corresponding 2-vinylallyamines in good yields (Table 3, entries 3–5). Nevertheless, the reaction of 2,3-pyridyne gave a

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complex mixture and the desired rearrangement product was not isolated (Table 3, entry 6).

Conclusions

In summary, we have established an unprecedented [2,3]-sigmatropic rearrangement reaction of quaternary 2,3-allenylammonium ylides, generated *in situ via* nucleophilic addition of tertiary 2,3-allenylamines to arynes, under mild conditions. Using 2-(trimethylsilyl)aryl triflates as aryne precursors, a range of tertiary 2,3-allenylamines bearing an electron-withdrawing group at the α-position smoothly participated in the aryne-mediated [2,3]-sigmatropic rearrangement at room temperature, delivering structurally diverse 2-vinylallyamines or 1-amino-1,3-dienes in moderate to excellent yields. The reaction proceeds in the absence of strong bases and transition metals, is compatible with moisture and air, and tolerates a wide variety of functional groups.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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Notes and references

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